

Chapter 3. **GROUPS**

3.4 Homomorphisms and isomorphisms



Today we will

- talk about non-isomorphic groups
- Isomorphism between cyclic groups

How can we show that two groups are not isomorphic?

If a group G_1 has a property and G_2 does not have it, or vice versa, we can conclude that G_1 and G_2 are not isomorphic.

Some properties to check out 😊

1. G_1 is commutative, and G_2 is not.
2. G_1 is cyclic, and G_2 is not.
3. G_1 is generated by two elements and G_2 is not generated by any choice of two of its elements.
4. Every element of G_1 is the square of an element of G_1 , whereas G_2 does not have this property.

Claim. The additive group $(\mathbb{R}, +)$ is not isomorphic to the multiplicative group $(\mathbb{R}^*, \cdot)$.

Notice that in $\mathbb{R}^*$ we have that

$$(-1)(-1) = (-1)^2 = 1 \quad \text{and} \quad \underline{-1 \neq 1}.$$

Note that 1 is the identity element of $\mathbb{R}^*$.

If $\varphi: (\mathbb{R}^*, \cdot) \rightarrow (\mathbb{R}, +)$ is a group isomorphism, we have that $\varphi(1) = 0$.

Moreover, $\varphi(-1) \neq 0$.

$$\varphi((-1)(-1)) = \underline{\varphi(-1) + \varphi(-1)}$$

$$\underline{\varphi(1) = 0}$$

We have obtained a nonzero real number $a = \varphi(-1)$ such that

$$a + a = 0.$$

But this is impossible, so the groups $(\mathbb{R}^*, \cdot)$ and $(\mathbb{R}, +)$ are not isomorphic.

THEOREM 3.42. ISOMORPHISM OF CYCLIC GROUPS.

- (1) For every positive integer n , every cyclic group of order n is isomorphic to $\mathbb{Z}_n$. Thus, any two cyclic groups of order n are isomorphic.
- (2) Every cyclic group of order infinity is isomorphic to $\mathbb{Z}$, and therefore any two cyclic groups of order infinity are isomorphic.

Proof.

(1) Let $C_n = \langle a \rangle$ be a (multiplicative) cyclic group of order n . Recall that

$$C_n = \langle a \rangle = \{1, a, a^2, \dots, a^{n-1}\}.$$

On the other hand, we have

$$\mathbb{Z}_n = \{ \bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1} \}.$$

$$C_n = \langle a \rangle = \left\{ \overset{a^0}{1}, a^1, a^2, \dots, a^{n-1} \right\}$$

$$\mathbb{Z}_n = \{ \bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1} \}$$

Let $\varphi: \mathbb{Z}_n \longrightarrow C_n$ given by

$\varphi(\bar{i}) = a^i$. It is clearly bijective.

$$\varphi(\bar{i} + \bar{j}) \stackrel{?}{=} \varphi(\bar{i})\varphi(\bar{j}), \text{ for all } \bar{i}, \bar{j} \in \mathbb{Z}_n.$$

$$\begin{aligned} \varphi(\bar{i} + \bar{j}) &= \varphi(\overline{i+j}) = a^{i+j} = a^i \cdot a^j \\ &= \varphi(\bar{i})\varphi(\bar{j}) \end{aligned}$$

② Suppose that $\langle a \rangle$ is a cyclic group of infinite order.

$$\begin{array}{ccccccc} \langle a \rangle = \{ \dots, a^{-2}, a^{-1}, \overset{a^0}{1}, a^1, a^2, \dots \} \\ \updownarrow \quad \updownarrow \quad \updownarrow \quad \updownarrow \quad \updownarrow \\ \mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \} \end{array}$$

The map $\varphi: \mathbb{Z} \rightarrow \langle a \rangle$ given by $\varphi(i) = a^i$ for all $i \in \mathbb{Z}$ is a group isomorphism. $\blacksquare$